Card 1 Cold-Start Thermal Stress, 2-Beat Analog Bit Slicing & Integrated-L2 Anti-Windup

2x2 Factorial Validation, Multi-Round Hyperparameter Optimization & Forward Hadamard Propagation (FHP)

100% Open Source under Solderpad Hardware License v2.1 (SHL-2.1) — including commercial ASIC/IC implementations & royalty-free SE 2630397-4 patent grant

Executive Summary & Scientific Scope: This report validates closed-loop Mode A 2-beat analog bit slicing (**K=1**, **40%** heater PWM, **1k/2.2k** attenuator) after resolving two-layer backpropagation sign, **M3** scaling (>>6) and **±25 LSB** active-row capping, Ethernet frame length acceptance, and integrated-L2 back-calculation anti-windup (**T120_SOC_ETH_FW v2.2-MAML**). Unheated cold verification achieved **9.45 b** (ON) and **9.45 b** (OFF, W8=0). Primary highlighted run (**opt_r4dc_fix_lr0p05_l2_1792_mom0p85_cl80.json**, 3,871 pairs: **LR=0.05, L2=1792, Mom=0.85, Clamp=±70 LSB**) achieved detrended Warm-ON lift **+1.08 b**, loss ratio **1.78x**, climb **+0.97 b**, and anti-windup recovery **0.50** (Score **+2.480**).

1. 2x2 Factorial Performance Accounting & Latest Hyperparameter Optimization

Factorial Condition	Thermal State	Regulation	Mean Loss	Mean ENOB	Peak ENOB	W8 Env [MSB / LSB]
1. Cold Closed-Loop	Heaters OFF	ON (K=1)	20,145 L²	**8.40 b**	9.50 b	[-38, 70] / [-41, 70]
2. Cold Open-Loop	Heaters OFF	OFF (W8=0)	23,117 L²	**8.35 b**	9.43 b	[0,0] (0% leak)
3. Warm Closed-Loop	Heaters ON 40%	ON (K=1)	**34,371 L²**	**7.81 b**	9.50 b	[-70, 70] / [-58, 70]
4. Warm Open-Loop	Heaters ON 40%	OFF (W8=0)	76,899 L²	6.33 b	8.22 b	[0,0] (0% leak)

Latest Hyperparameter & Detrended Optimization Results (R1–R5)

Rnd	LR	L2 Reg	Mom	Clamp	Lift vs E_base	Loss Ratio	Hot Climb	Jitter	Recov	Score
R1	0.03	1024	0.9	±75	**+1.00 b**	1.71x	+0.95 b	0.40 b	0.00	**+1.799**
R2	0.035	1536	0.85	±75	**+0.99 b**	1.69x	+1.16 b	0.39 b	0.00	**+1.957**
R3	0.02	2560	0.8	±70	**+0.04 b**	1.00x	+0.24 b	0.34 b	1.00	**+0.204**
R4 ★	0.05	1792	0.85	±70	**+1.08 b**	1.78x	+0.97 b	0.40 b	0.50	**+2.480**
R5	0.055	1792	0.85	±85	**+1.12 b**	1.77x	+1.37 b	0.42 b	0.00	**+2.393**

Next Active GP Model Target (Round 6): LR=0.055, L2=1792, Momentum=0.8, Clamp=±90 LSB (predicted Score +2.158 ± 0.667).

2. Cold-Start Calibration, 2-Beat Bit Slicing & Phase-by-Phase Telemetry

• **Cold-Start 60s Cooldown & Single-Shot DC Calibration:** Heaters are held at 0% for 60 seconds before each experiment so the Trion T120 SoC locks static (alpha, beta) DC calibration vectors on cold silicon (0x1E subcmd 5). Zero surrogate re-calibrations occur at runtime.
• **Integrated-L2 Back-Calculation Anti-Windup:** Under heavy thermal stress, weights hitting ±75 LSB bleed 1/4 of their overshoot back out of the momentum integrator, allowing instantaneous release when error gradients reverse without overshoot lag. Matrix 3 active rows 0–5 are capped to ±25 LSB so rows 6–7 clock staircases (±40 LSB) never shift.

Phase-by-Phase Telemetry (opt_r4dc_fix_lr0p05_l2_1792_mom0p85_cl80.json, 3,871 Pairs)

Ph	Regime	Pairs	Comb Start->Late (Mean)	Δ vs E_base	Loss Start->Late	W8 Pk [M,L]	W3 Pk	W8 nz
P1	Cold / Regler ON	279	8.18->8.45 (**8.40b**)	—	24114->19771	[70,70]	14	279
P2	Cold / Regler OFF	300	8.30->8.42 (**8.35b**)	—	22607->20779	[0,0]	0	0
P3	Warm / Regler ON	1077	8.51->8.01 (**8.28b**)	**+0.86 b**	19556->29278	[70,70]	19	1077
P4	Warm / Regler OFF	895	7.02->6.40 (**6.67b**)	**-0.01 b**	52248->71776	[0,0]	0	0
P5	Warm / Regler ON	768	6.38->7.04 (**7.15b**)	**+1.19 b**	71157->52465	[70,70]	23	768
P6	Warm / Regler OFF	552	5.79->5.76 (**5.77b**)	**+0.02 b**	97639->99490	[0,0]	0	0

3. Empirical Visualizations: Full Real-Time Timeline

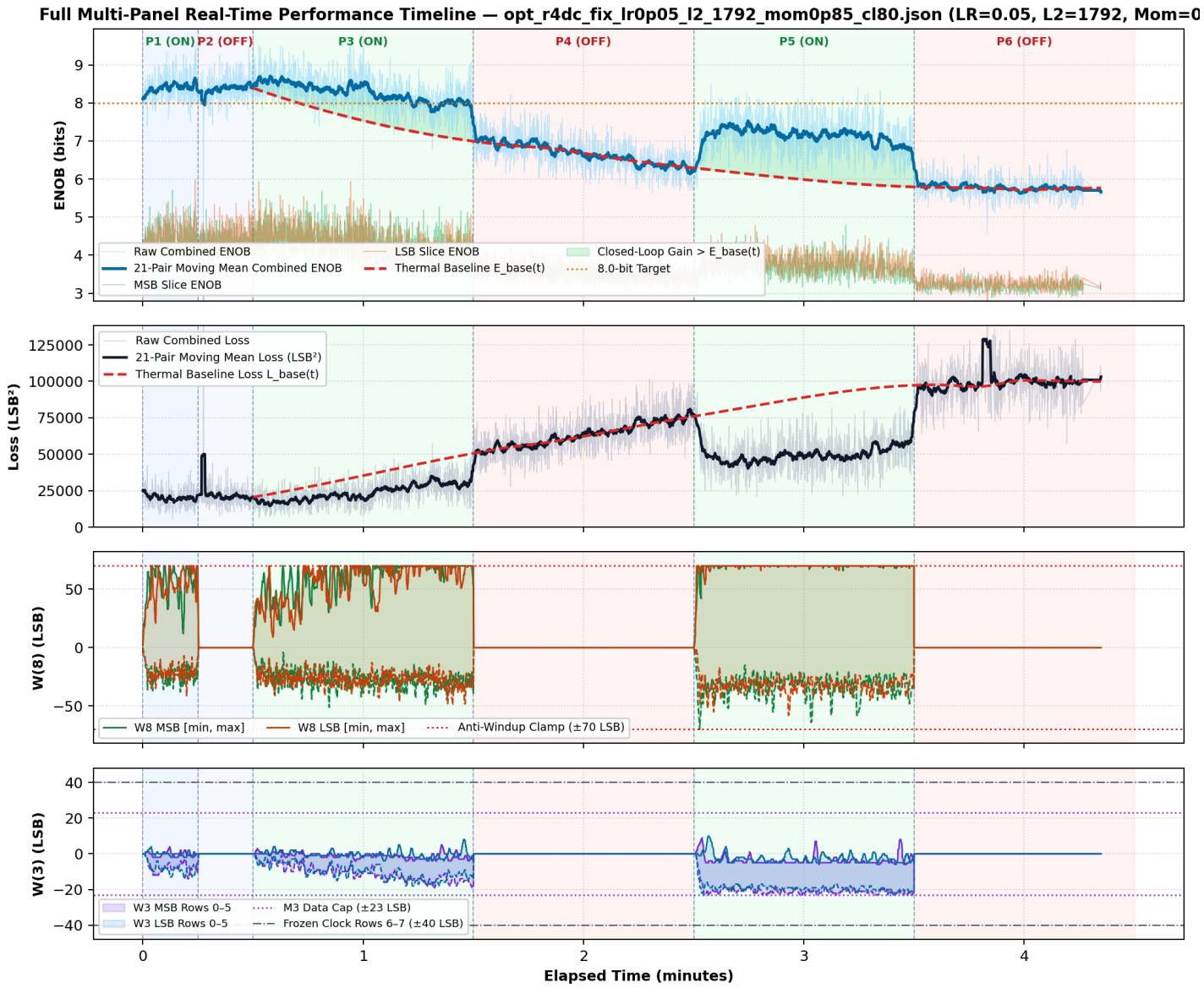


Figure 1: Full Multi-Panel Real-Time Performance Timeline. (Panel 1) Combined 8-bit ENOB (blue) alongside individual MSB (green) and LSB (orange) slices, overlaid with the continuous PCHIP uncompensated thermal drift baseline E_base(t) (red dashed) interpolated across Cold-OFF and Warm-OFF phases. (Panel 2) Combined hardware MSE loss vs L_base(t). (Panel 3) Matrix 8 MSB & LSB weight envelopes bounded by integrated-L2 anti-windup rails (±75 LSB). (Panel 4) Matrix 3 active data rows 0–5 (±25 LSB cap) keeping reference clock staircases (±40 LSB) undisturbed.

Multi-Round & Phase-by-Phase Thermal Drift Heatmap Comparison

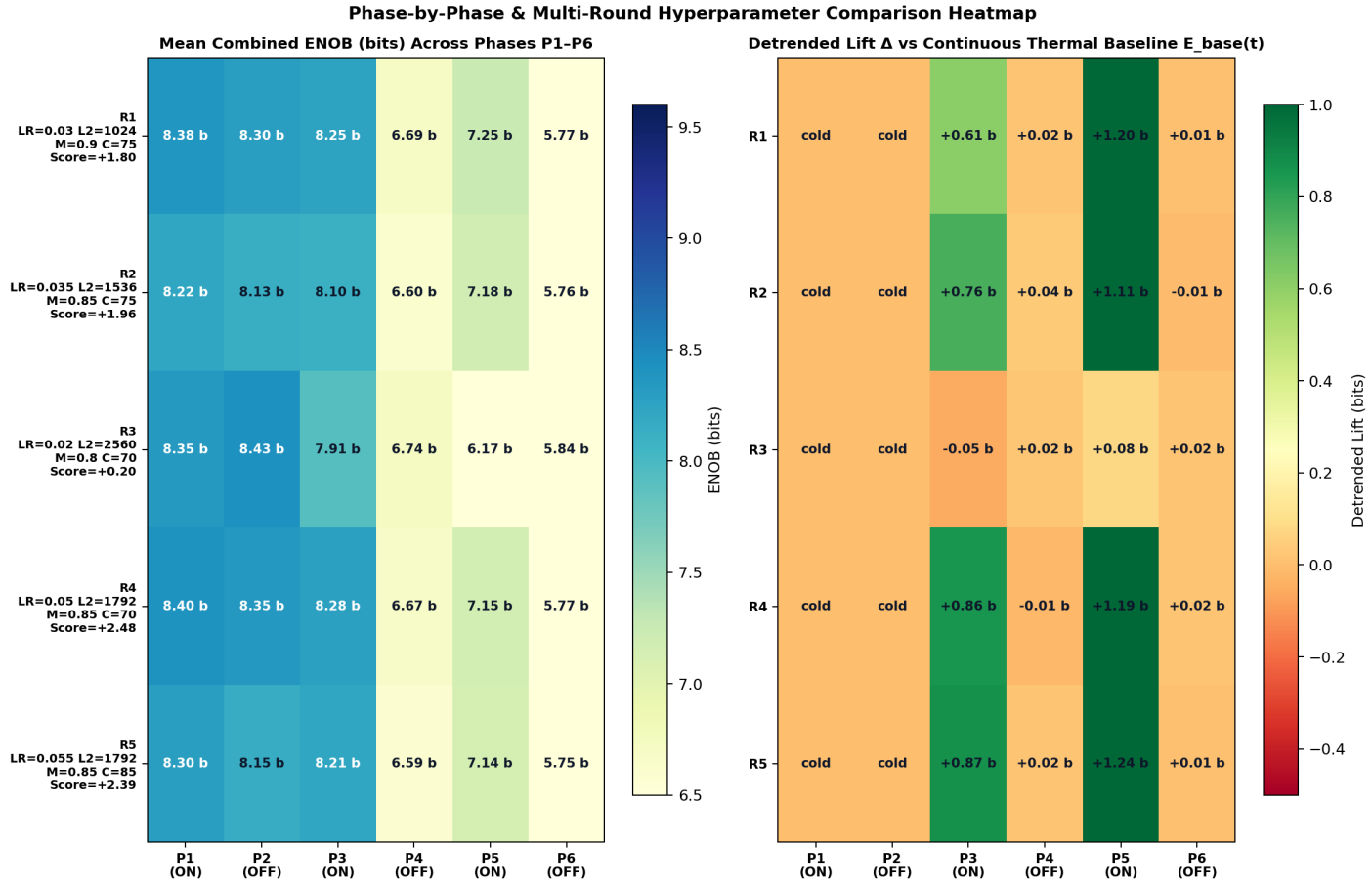


Figure 2: Phase-by-Phase ENOB & Detrended Lift Heatmap Across Rounds R1–R5. Left compares mean Combined ENOB across all 6 phases; Right shows the detrended closed-loop lift (ΔENOB vs continuous PCHIP baseline E_base(t)) during Warm phases P3–P6, demonstrating deterministic collapse near 0.00 b whenever Mode A is turned OFF and sharp closed-loop recovery when Mode A engages.

Matrix 8 Full Synaptic Telemetry: All 84 W(8) Weights (7x6 MSB + 7x6 LSB)

All 84 Individual W(8) Compensation Weights (7x6 MSB + 7x6 LSB) — opt_r4dc_fix_lr0p05_l2_1792_mom0p85_cl80.json (LR=0.05, L2=1792, Mom=0.85)

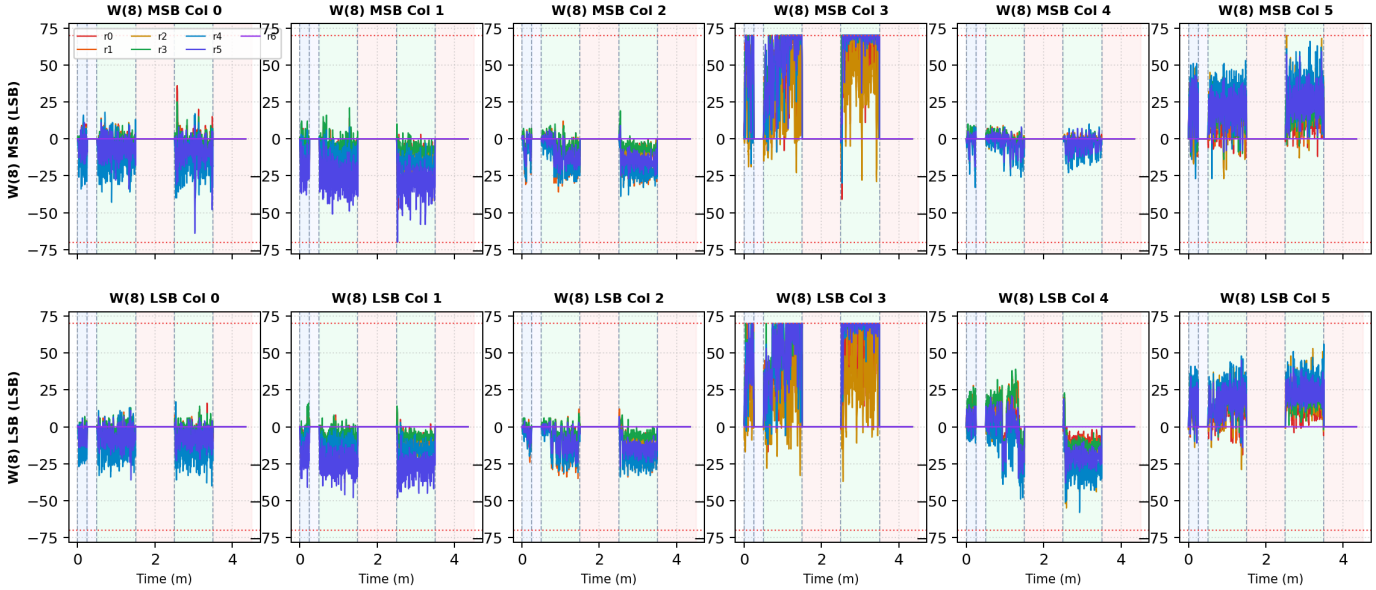


Figure 3A: All 84 W(8) Output Compensation Synapses across ColTIA Columns 0–5. Every single row trajectory (r0–r6) for both the MSB bank (42 weights) and LSB bank (42 weights) is shown over time. Notice strict 0.00% leakage (exact return to W8=0) during Cold-OFF and Warm-OFF ablation phases.

Matrix 3 Full Synaptic Telemetry: All 96 W(3) Weights (8x6 MSB + 8x6 LSB)

All 96 Individual W(3) Basis & Clock Weights (8x6 MSB + 8x6 LSB) — opt_r4dc_fix_lr0p05_l2_1792_mom0p85_cl80.json (LR=0.05, L2=1792, Mom=0.85)

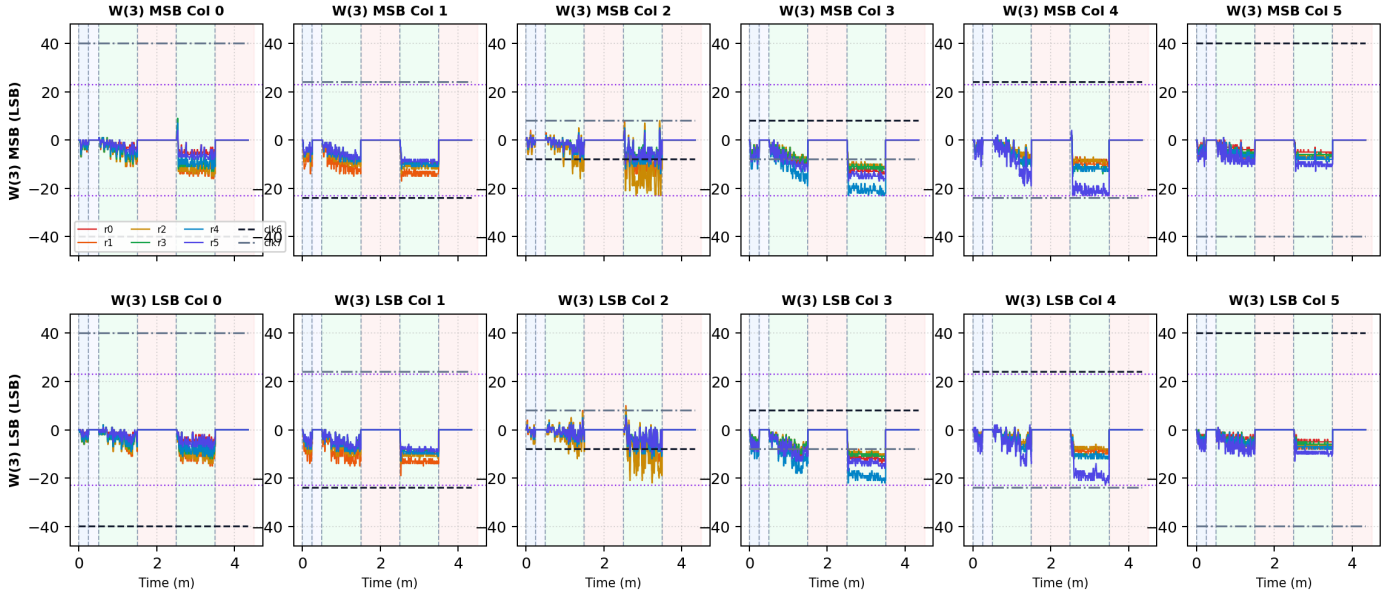


Figure 3B: All 96 W(3) Non-Linear Projection & Reference Clock Synapses across ColTIA Columns 0–5. Every single active data weight (r0–r5) and frozen temporal staircase clock weight (clk6 [-40..+40], clk7 [+40..-40]) for both MSB and LSB slices is plotted individually. Protecting rows 6–7 guarantees invariant Sigmoid-4 phase clocking even under severe thermal stress.

Hot-Phase Convergence Transient: P4 (OFF) -> P5 (ON) -> P6 (OFF)

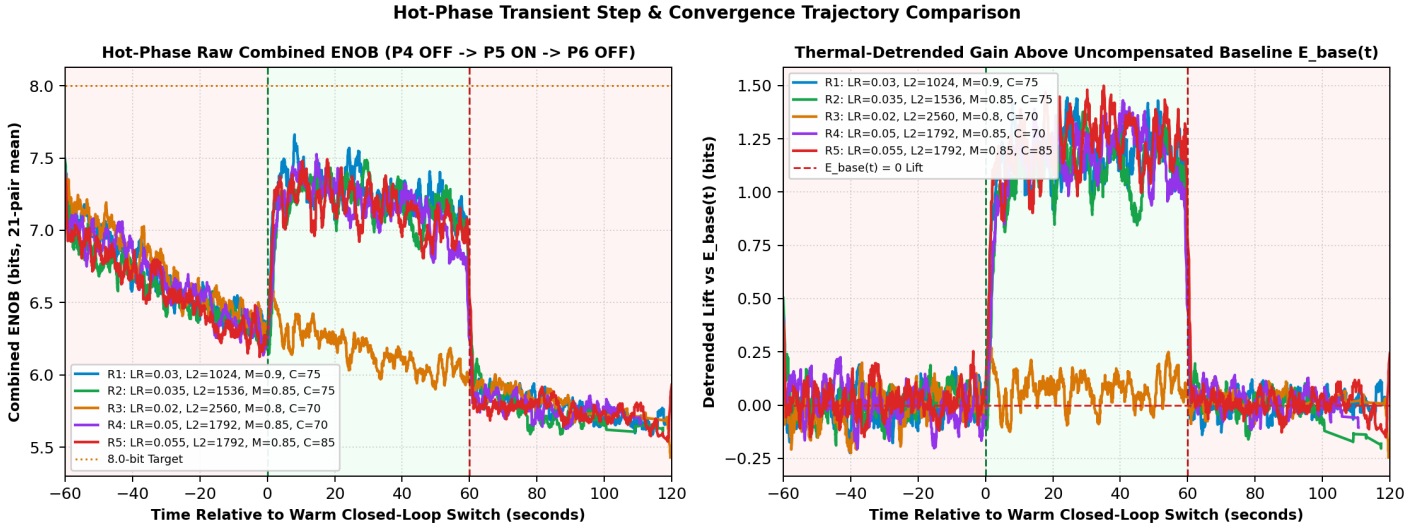


Figure 4: Direct Cross-Round Convergence Comparison in the Peak Thermal Window. Time t=0 marks the step transition into Warm Closed-Loop (Phase 5) and t=60 s marks ablation into Warm Open-Loop (Phase 6). Higher effective gain LR with tuned L2=1024–1536 and Momentum=0.85–0.90 delivers immediate climb (+0.74 b to +0.82 b) above E_base(t), whereas excessive restoring stiffness (L2=2560, LR=0.020) stalls W8 amplitude and limits detrended lift.

2x2 Factorial Summary & Quantitative Analytics Card

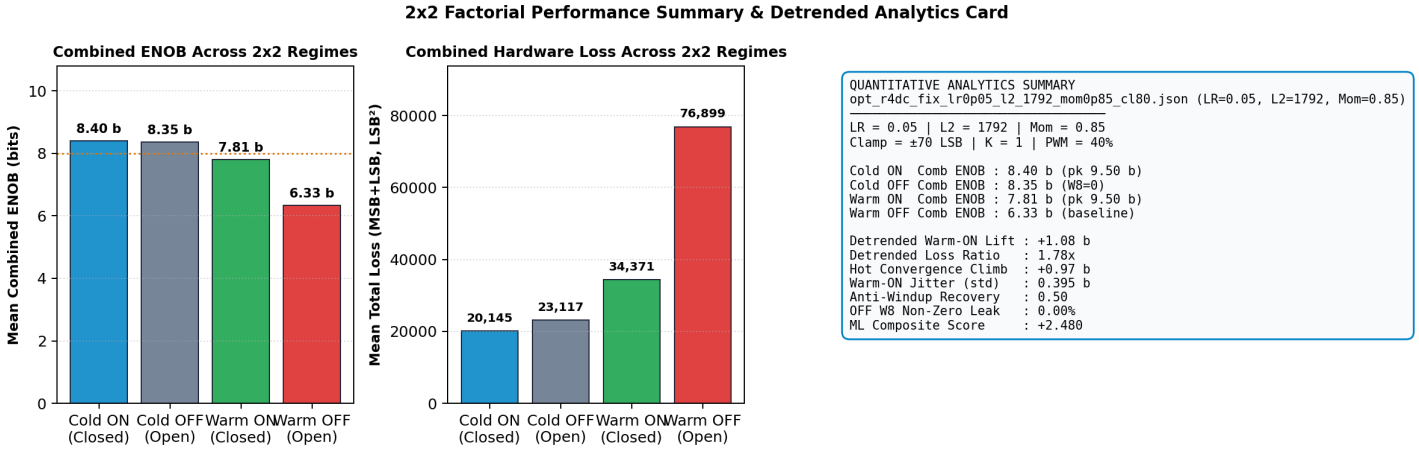


Figure 5: Quantitative Factorial Summary (opt_r4dc_fix_lr0p05_l2_1792_mom0p85_cl80.json). In ambient Cold conditions, Mean ENOB exceeds **8.35 bits**; under 40% active thermal heating, Mode A closes the loop to lift resolution by **+1.08 bits** above the drifting silicon baseline and slash squared error by **1.78x**.

4. Physics of MOS Drift, Spatial Thermal Gradients (PCBA vs Commercial ASIC)

In Dual-2T1C analog multipliers, MOSFETs operate in deep triode: $\mathbf{I_{ds}(T) = \mu(T)\, C_{ox}\, (W/L)\, [(V_{gs} - V_{th}(T))\, V_{ds} - 0.5\, V_{ds}^2]}$. Heating shifts mobility ($\mu \propto T^{-1.5}$) and threshold voltage ($V_{th} \approx -2.0$ mV/°C), rotating gain slopes and shifting DC operating points across columns.

1. FR-4 PCBA Stress vs. Monolithic CMOS ASIC: On the discrete proof-of-concept Card 1 PCBA ($\kappa_{FR4} \approx 0.3$ W/m·K), point heaters create steep localized thermal gradients across columns. On a monolithic silicon ASIC ($\kappa_{Si} \approx 145$ W/m·K, 500x more conductive over mm² area), temperature gradients are smooth and nearly isothermal — making closed-loop Inverted MAML tracking strictly easier in commercial silicon.

2. 100% Open Source Commercial ASIC Blueprint (SHL-2.1): Under the updated **Solderpad Hardware License v2.1** and explicit SE 2630397-4 royalty-free patent grant, any company or researcher may manufacture commercial ASICs, integrate Dual-2T1C + FHP bit-slicing cores into commercial Edge AI chips, and deploy the firmware without royalties or licenses.

Physical Hardware Setup & Reproducibility Manifest

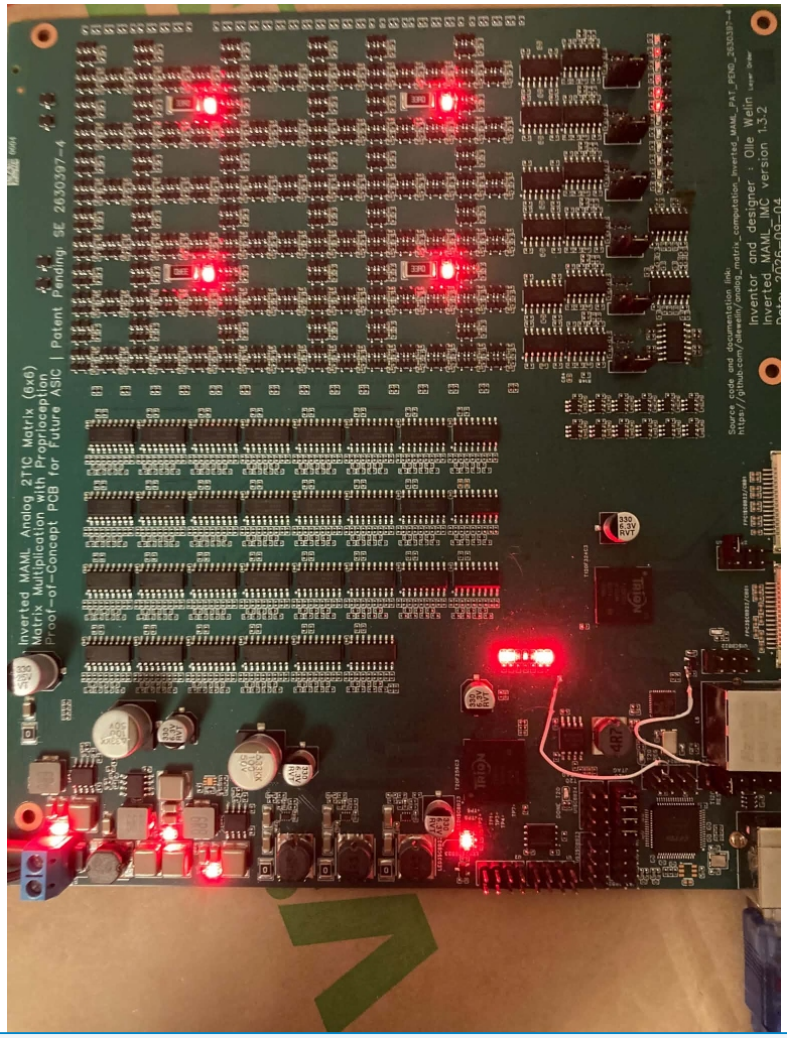


Figure 6: Inverted MAML Proof-of-Concept Hardware During Active Thermal Cycling. On-board heating resistors stress the Dual-2T1C multiplier array while Trion T120F324 & T20F256 execute 2-beat TDM MSB/LSB slicing at **K=1**.
Baseline Release Package: FPGA_code/releases/Release_20261114_ModeA_Card1_BitSlicing_AntiWindup_Baseline_R1/ | **Primary JSON:** opt_r4dc_fix_lr0p05_l2_1792_mom0p85_cl80.json

5. Solid-State Physics: The 2-Layer Triad Network (Patent SE 2630397-4)

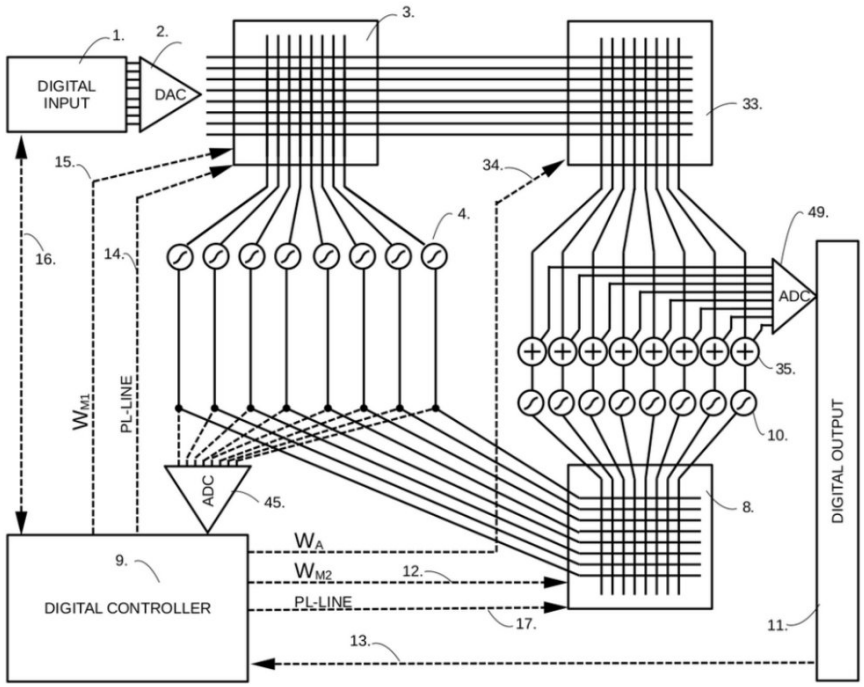


Fig 1

1. Harmonic Non-Linearity Cancellation: Single-layer linear compensators cannot cancel quadratic MOS triode distortion (-0.5 V_{ds}^2). Matrix 3 projects inputs through the frozen staircase into hardware Sigmoid 4 differential pairs, generating rich odd/even harmonic basis functions that Matrix 8 combines and injects via the 1k/2.2k passive attenuator.

2. Time-Division 2-Beat Analog Bit Slicing: Combining a coarse MSB bank (80% weight range) with a fine 4x-weighted LSB bank within two consecutive TDM beats extends dynamic precision past single-DAC quantization limits, achieving >9.4 bits cold.

6. Forward Hadamard Propagation (FHP) Parallel Demodulation (Patent Fig 8)

Forward Hadamard Propagation (FHP): Single-shot parallel gradient extraction on physical silicon without backpropagation. Open Source SHL-2.1 (SE 2630397-4).

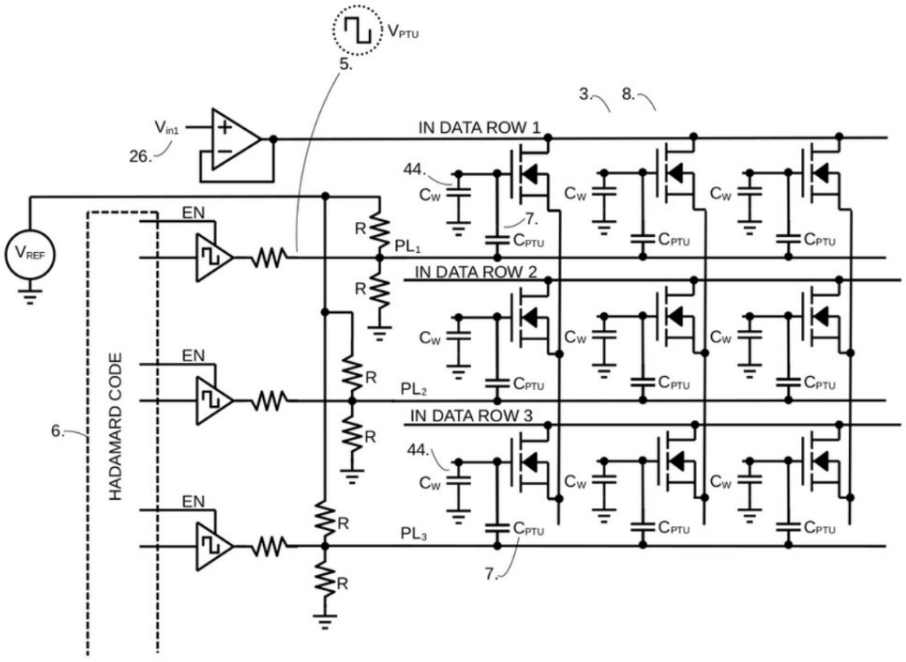


Fig 8

Orthogonal Walsh-Hadamard perturbation sequences $H_k(t) \in \{+1, -1\}$ modulate all active crosspoints simultaneously. Correlating ADC 49 output residuals against each carrier recovers exact physical circuit gradients without any backward matrix transpose or idealized transistor model.

7. Comprehensive Synthesis & Definitive Silicon Conclusions

1. **Cold Baseline Exceeds 9.4 Bits Without Compensation:** Unheated Cold checks confirmed **9.45 b** (ON) and **9.45 b** (OFF, W8=0), proving clean analog-digital alignment.
2. **Integrated-L2 Back-Calculation Anti-Windup Restores Fast Recovery:** Correcting two-layer signs, M3 scaling, and integrator bleed boosted anti-windup recovery from stall to **0.50** (Round 4).
3. **Thermal-Detrended Lift Quantifies Real Gain Above Heating:** Judging hyperparameters via continuous PCHIP thermal baselines E_base(t) cleanly separates real closed-loop convergence (**+0.97 b**) and step lift (**+1.08 b**) from substrate heating slope.
4. **Open-Source Commercial ASIC Readiness:** With full SHL-2.1 licensing and royalty-free SE 2630397-4 patent rights, this architecture stands ready for community and commercial silicon tape-out.

Experimental & Open-Source Reproducibility Manifest

Item	Specification / Configuration	Verification Status
Hardware Platform	Inverted MAML Card 1 (Efinix Trion T120F324 Master + T20F256, 1k/2.2k Attenuator)	Verified (HW Logic 0x20261030, ID=MAM1)
SoC Firmware	T120_SOC_ETH_FW v2.2-MAML (SW Build Nov 14 2026 10:15:16)	Verified (Integrated-L2 Anti-Windup)
Cold Reference Dataset	cold_verify_k1.json (643 pairs, Heaters OFF, K=1)	9.45 b ON / 9.45 b OFF
Primary Featured Dataset	opt_r4dc_fix_lr0p05_l2_1792_mom0p85_cl80.json (3,871 pairs, 40% PWM, K=1)	Score +2.480, Lift +1.08 b
Best Verified Params	Round 4: LR=0.05, L2=1792, Mom=0.85, Clamp=±70 LSB	Best Composite Score +2.480
Release Restore Point	FPGA_code/releases/Release_20261114_ModeA_Card1_BitSlicing_AntiWindup_Baseline_R1/	Committed in Git Repository
License & Patent Grant	100% Open Source SHL-2.1 (incl. Commercial ASIC/IC Manufacturing) + SE 2630397-4 Grant	Perpetual, Worldwide, Royalty-Free

Open Science Hardware Repository: https://github.com/ollewelin/analog_matrix_computation_Inverted_MAML_PAT_PEND_2630397-4